

Cabe Robertson

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OBJECTIVE

Pursuing Electrical Engineering internship roles in hardware design, test, and validation for Summer 2027

EDUCATION

New Mexico State University - Las Cruces, NM

Expected December 2027

- Bachelor of Science in Electrical Engineering (Communications and Signal Processing Concentration)

SKILLS

HARDWARE: Oscilloscope, Logic Analyzer, Spectrum Analyzer, Function Generator, Power Supply, Anemometer, Microcontrollers

SOFTWARE: Arduino IDE, C, C++, Python, MATLAB, EAGLE, VHDL, Quartus Prime, Flask, Git, AutoCAD, Overleaf

LANGUAGE: English (Native), Spanish (Native)

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern, Emerging Technology Ventures (ETV America) - Las Cruces, NM

Jul 2026 - Present

- Designed the equipment layout for the company's unmanned systems production floor, specifying tooling and power requirements to support a 7 person build team
- Qualified vendors and handled procurement for the entire floor, delivering a turnkey build area ready for staff move-in
- Formulated a 59 line item bill of materials totaling roughly \$168,000 to source every component through Blue UAS and NDAA compliant vendors
- Co-authored a 30 page NASA SBIR/STTR proposal pursuing a \$2.5 million matching award for an autonomous on-orbit inspection and repair system

Electrical Engineering Intern, GILZ LLC - Remote

Mar 2026 - May 2026

- Deconstructed a desk blower fan to test the internal BLDC motor's voltage/current ratings and PWM capabilities for use in an air filtration system
- Drafted an Arduino program on an ESP32 to control the 3 speeds of the BLDC motor to add user control capabilities
- Conducted benchmarks with the original blower fan to measure noise, temperature, speed, and velocity using different attachments such as air ducts and filters, isolating the configuration that cut noise by roughly 10 dB while raising airflow 20%

Hardware Engineering Intern, Syndetix Incorporated - Las Cruces, NM

Apr 2025 - Aug 2025

- Created an affordable demonstration kit for \$219 to accurately present the capabilities of a communication disrupting device to law enforcement and defense agencies while also being easily producible
- Improvised a lightweight solution for a scenario in which wind could knock over the demonstration, ensuring portability and stability of the kit
- Formulated an in-depth user manual for a new product, guiding customers on proper handling and set up of the device for missions or training
- Repaired a critical permission-based synchronization issue on the company server that had been active for 4 years, improving efficiency of workstations company wide

Research Assistant, Mitchell Coding Group - Las Cruces, NM

Aug 2025 - Feb 2026

- Developed Python programs to simulate channel decoding using Linear Algebra for wireless communication models
- Executed experiments with multiple hour runtimes utilizing contrasting bit string values to ensure software functionality
- Wrote MATLAB scripts to process and visualize experimental results, producing publication-quality plots and diagrams

ACADEMIC PROJECTS

Low-Cost EEG Electrode for Brain-Computer Interfaces, (NeuroPawn Biotechnologies)

Team Lead, 5-Person Team

- Initiated the partnership with NeuroPawn Biotechnologies and authored the project proposal that established it as an approved senior capstone, along with the IRB protocol governing human subjects testing
- Lead a 5 person team spanning electrical, electronics, mechanical, and mechanical engineering technology disciplines
- Designing and bench testing a low-cost dry EEG electrode, benchmarking contact impedence, contact area, DC offset, noise, frequency response, shielding, and durability against the incumbent commercial electrode
- Building a proof-of-concept brain-computer interface that uses the electrodes to steer a remote-control car through a maze, comparing performance between the new and incumbent designs

CrashGuard, (ESP32, MPU6050, Python, Flask)

IEEE MYOSA 6.0 Finalist

- Selected as 1 of 15 teams worldwide for the IEEE Sensors Council's international MYOSA 6.0 competition and advanced as 1 of 7 finalists, presenting a live hardware demonstration at IEEE SENSORS 2026 in Rotterdam
- Programmed the ESP32 and MPU6050 sensor firmware to flag vehicle impacts from 6-axis IMU data at a 50 g threshold
- Traced a fault to the vendor sensor library clipping crash level readings, then wrote direct I2C burst reads and corrected the accelerometer scale factor to recover the full data range
- Constructed a 46 test automated suite across BLE, a Flask call server, and a Twilio voice interface to verify the detection chain and eliminate false triggers

LEADERSHIP

Oct 2024 - Present

Vice President, IEEE NMSU Chapter

- Directing a unique fundraising project with an estimated revenue of \$50,000 to revitalize the chapter and ECE department
- Grew active membership from 10 to over 30 students, leading the turnaround that earned NMSU's Most Improved Student Organization Award
- Oversee and manage all club projects ensuring tasks are being completed efficiently, and mentor students during technical workshops teaching proper troubleshooting methods and soldering techniques
- Diplomat for communication with industry representatives, external IEEE chapters, and other student organizations on campus for sponsorship proposals and collaborations